

Problem 1a: What Are We Even Doing? (The Confused Student's Guide)

ECO 310 - Understanding Compensated Demand

February 23, 2026

1 You're Not Alone - This IS Confusing!

Let me break this down from scratch. The problem asks for "compensated demand" which sounds scary, but it's actually asking a simple question once you understand what's going on.

2 Two Different Questions Economists Ask

There are TWO different optimization problems in microeconomics that look similar but ask opposite questions:

2.1 Question 1: Regular (Marshallian) Demand

The Normal Question

I have \$100. How do I spend it to be as happy as possible?

Maximize: Utility $u(x_1, x_2)$

Subject to: Budget constraint $p_1x_1 + p_2x_2 \leq m$

This is what you do when you go shopping: "I have this much money, let me get the best stuff I can afford."

2.2 Question 2: Compensated (Hicksian) Demand

The Backwards Question

I want to be THIS happy (utility level u^*). What's the cheapest way to get there?

Minimize: Expenditure $p_1x_1 + p_2x_2$

Subject to: Must reach utility $u(x_1, x_2) \geq u^*$

This is like: "I need to be satisfied at level 10. What's the minimum I need to spend?"

KEY INSIGHT

Problem 1a asks you to solve Question 2 (the backwards one). We're minimizing cost, not maximizing utility!

3 Concrete Example: Coffee vs Tea

Let me give you a real example to understand this.

3.1 Setup

Your preferences: You like coffee and tea equally. One cup of coffee makes you just as happy as one cup of tea. So your utility is:

$$u(\text{coffee}, \text{tea}) = \text{coffee} + \text{tea} \quad (1)$$

These are **perfect substitutes** - you'd trade them 1-for-1 and not care.

3.2 Current Situation

- Coffee costs \$3 per cup ($p_{\text{coffee}} = 3$)
- Tea costs \$5 per cup ($p_{\text{tea}} = 5$)
- You want to reach happiness level 10 ($u^* = 10$)

Question: What's the cheapest way to get 10 units of happiness?

3.3 Possible Choices

You need coffee + tea = 10. Here are some options:

Coffee	Tea	Total Utility	Total Cost
10	0	10	$3(10) + 5(0) = \$30$
8	2	10	$3(8) + 5(2) = \$34$
5	5	10	$3(5) + 5(5) = \$40$
2	8	10	$3(2) + 5(8) = \$46$
0	10	10	$3(0) + 5(10) = \$50$

Which is cheapest? Buy 10 coffees, 0 teas for \$30.

Why? Coffee is cheaper! Since you like them equally, always buy the cheaper one.

The Answer for This Example

Compensated demand: $(x_{\text{coffee}}^c, x_{\text{tea}}^c) = (10, 0)$

This means: "To reach utility 10 cheaply, buy 10 coffees and 0 teas."

4 What If Prices Change?

4.1 Example 2: Tea Gets Cheaper

Now suppose:

- Coffee costs \$5 per cup ($p_{\text{coffee}} = 5$)
- Tea costs \$2 per cup ($p_{\text{tea}} = 2$)

- You still want utility level 10 ($u^* = 10$)

Same table:

Coffee	Tea	Total Utility	Total Cost
10	0	10	$5(10) + 2(0) = \$50$
8	2	10	$5(8) + 2(2) = \$44$
5	5	10	$5(5) + 2(5) = \$35$
2	8	10	$5(2) + 2(8) = \$26$
0	10	10	$5(0) + 2(10) = \$20$

Which is cheapest? Buy 0 coffees, 10 teas for \$20.

Why? Now tea is cheaper! So buy only tea.

The Answer for Example 2

Compensated demand: $(x_{\text{coffee}}^c, x_{\text{tea}}^c) = (0, 10)$

This means: "To reach utility 10 cheaply, buy 0 coffees and 10 teas."

5 The General Pattern

Do you see the pattern?

- When coffee is cheaper: Buy all coffee, no tea
- When tea is cheaper: Buy all tea, no coffee
- When they cost the same: Any mix works (you're indifferent)

This makes sense! If you like them equally, always buy the cheaper one.

6 What Happens When Prices Are Equal? ($p_1 = p_2$)

This case confuses people, so let's really break it down.

6.1 Example: Both Goods Cost \$4

Suppose coffee and tea both cost \$4, and you want utility $u^* = 10$.

You need 10 total units (since $u = x_1 + x_2$). Let's check different combinations:

Coffee	Tea	Total Utility	Total Cost
10	0	10	$\$4(10) + \$4(0) = \$40$
8	2	10	$\$4(8) + \$4(2) = \$40$
5	5	10	$\$4(5) + \$4(5) = \$40$
3	7	10	$\$4(3) + \$4(7) = \$40$
0	10	10	$\$4(0) + \$4(10) = \$40$

Every combination costs exactly \$40!

6.2 Why Does This Happen?

Since both goods:

- Give 1 utility per unit
- Cost the same price ($p_1 = p_2 = p$)
- And you need $u^* = 10$ total units

Then any mix of 10 units costs: $p \times 10 = \$4 \times 10 = \40

Mathematical explanation:

$$\begin{aligned}\text{Cost} &= p_1 \cdot x_1 + p_2 \cdot x_2 \\ &= p \cdot x_1 + p \cdot x_2 \quad (\text{since } p_1 = p_2 = p) \\ &= p(x_1 + x_2) \\ &= p \cdot u^* \quad (\text{since } x_1 + x_2 = u^*)\end{aligned}$$

No matter how you split x_1 and x_2 , as long as they add to u^* , the cost is the same!

6.3 What Do You Write Down?

For compensated demand:

When $p_1 = p_2$, the answer is: **any** $(x_1, u^* - x_1)$ **where** $0 \leq x_1 \leq u^*$

This means:

- You can buy $(10, 0)$ (all good 1)
- You can buy $(7, 3)$ (7 of good 1, 3 of good 2)
- You can buy $(5, 5)$ (half and half)
- You can buy $(0, 10)$ (all good 2)
- Literally **any split** where the numbers add to $u^* = 10$

All are equally optimal!

For the expenditure function:

Even though demand has infinitely many solutions, expenditure is unique:

$$e(p_1, p_2, u^*) = p_1 \cdot u^* = p_2 \cdot u^* = \$4 \times 10 = \$40$$

Key Insight

When $p_1 = p_2$:

- **Demand is indeterminate** (infinitely many optimal bundles)
- **Expenditure is determinate** (unique cost)

This is because every optimal bundle costs the same amount!

7 Writing the Mathematical Answer

The problem asks you to write this as a general formula. Let's use x_1 for good 1 and x_2 for good 2 (instead of coffee and tea).

Given:

- Utility function: $u(x_1, x_2) = x_1 + x_2$
- Prices: p_1 and p_2
- Target utility: u^*

Compensated demand formula:

Answer to Problem 1a

$$(x_1^c(p_1, p_2, u^*), x_2^c(p_1, p_2, u^*)) = \begin{cases} (u^*, 0) & \text{if } p_1 < p_2 \text{ (good 1 is cheaper)} \\ (0, u^*) & \text{if } p_1 > p_2 \text{ (good 2 is cheaper)} \\ \text{any } (x_1, u^* - x_1) \text{ where } 0 \leq x_1 \leq u^* & \text{if } p_1 = p_2 \end{cases} \quad (2)$$

Translation:

- If good 1 is cheaper ($p_1 < p_2$): Buy u^* units of good 1, and 0 of good 2
- If good 2 is cheaper ($p_1 > p_2$): Buy 0 of good 1, and u^* units of good 2
- If they cost the same ($p_1 = p_2$): Any combination that adds to u^* works

8 Numerical Check (Using Coffee Example)

Let's verify our formula works:

Example 1: $p_{\text{coffee}} = 3$, $p_{\text{tea}} = 5$, $u^* = 10$

Since $3 < 5$, coffee is cheaper. Our formula says: $(u^*, 0) = (10, 0)$

Check: 10 coffees + 0 teas. Utility = $10 + 0 = 10$. Cost = $3(10) + 5(0) = \$30$ (cheapest option).

Example 2: $p_{\text{coffee}} = 5$, $p_{\text{tea}} = 2$, $u^* = 10$

Since $5 > 2$, tea is cheaper. Our formula says: $(0, u^*) = (0, 10)$

Check: 0 coffees + 10 teas. Utility = $0 + 10 = 10$. Cost = $5(0) + 2(10) = \$20$ (cheapest option).

9 Why Is This Called "Compensated" Demand?

The name is confusing, but here's the idea:

- In regular demand: "You have \$100, prices change, so you adjust what you buy."
- In compensated demand: "Prices change, but we imagine giving you just enough money (compensating you) so you can still afford the same happiness level."

The "compensation" part means we're holding utility constant at u^* , not holding money constant.

10 Why Do Economists Care About This?

Short answer: To separate price effects into two parts:

1. **Substitution effect:** How do you change consumption when price changes, holding utility constant? (This uses compensated demand!)
2. **Income effect:** How does the change in your purchasing power affect consumption?

Parts (c)i and (c)ii of the problem ask you to calculate substitution effects, which use your answer to part (a).

11 Summary: What You Need to Remember

TL;DR

Compensated demand asks: What's the cheapest way to reach utility level u^* ?

For perfect substitutes ($u = x_1 + x_2$): Always buy only the cheaper good.

Your answer:

$$(x_1^c, x_2^c) = \begin{cases} (u^*, 0) & \text{if } p_1 < p_2 \\ (0, u^*) & \text{if } p_1 > p_2 \\ \text{any mix adding to } u^* & \text{if } p_1 = p_2 \end{cases}$$

Intuition: If goods are perfect substitutes and you want to minimize cost, buy the cheap one!

12 Connection to Part (b): Expenditure Function

12.1 What IS the Expenditure Function?

The expenditure function answers this question:

"What's the MINIMUM amount of money I need to reach utility level u^ ?"*

It's literally asking: **How much do I have to spend?**

12.2 Building the Expenditure Function Step-by-Step

Let's build this from scratch using our coffee/tea example.

Setup:

- Coffee costs \$3, tea costs \$5
- You want utility level 10 (meaning $u^* = 10$)
- From part (a), we found compensated demand: $(x_1^c, x_2^c) = (10, 0)$
- Translation: Buy 10 coffees, 0 teas

Question: How much does this shopping cart cost?

Answer:

$$\begin{aligned}\text{Total cost} &= (\text{price of coffee}) \times (\text{coffees bought}) + (\text{price of tea}) \times (\text{teas bought}) \\ &= \$3 \times 10 + \$5 \times 0 \\ &= \$30 + \$0 \\ &= \$30\end{aligned}$$

That's it! The expenditure function is just the **total bill** for your optimal shopping cart.

12.3 Writing It as a Formula

In general notation:

$$e(p_1, p_2, u^*) = p_1 \cdot x_1^c + p_2 \cdot x_2^c \quad (3)$$

Translation:

- e = expenditure (the total cost)
- $p_1 \cdot x_1^c$ = (price of good 1) $\times$ (optimal quantity of good 1)
- $p_2 \cdot x_2^c$ = (price of good 2) $\times$ (optimal quantity of good 2)

Key insight: We're using the quantities from compensated demand (x_1^c, x_2^c) , which we already found in part (a)!

12.4 Plugging In Our Answer

From part (a), we know:

- If $p_1 < p_2$: $(x_1^c, x_2^c) = (u^*, 0)$ (buy only good 1)
- If $p_1 > p_2$: $(x_1^c, x_2^c) = (0, u^*)$ (buy only good 2)

So the expenditure is:

$$e(p_1, p_2, u^*) = p_1 \cdot x_1^c + p_2 \cdot x_2^c \quad (4)$$

$$= \begin{cases} p_1 \cdot u^* + p_2 \cdot 0 = p_1 u^* & \text{if } p_1 < p_2 \\ p_1 \cdot 0 + p_2 \cdot u^* = p_2 u^* & \text{if } p_1 > p_2 \\ p_1 u^* = p_2 u^* & \text{if } p_1 = p_2 \end{cases} \quad (5)$$

$$= \min\{p_1, p_2\} \cdot u^* \quad (6)$$

12.5 What Does This Mean in English?

The final formula $e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^*$ says:

Minimum cost = (cheaper price) $\times$ (target utility)

Why does this make sense?

Since goods are perfect substitutes:

1. You need exactly u^* total units to reach utility u^* (because $u = x_1 + x_2$)
2. You'll buy all u^* units of whichever good is cheaper
3. So your cost is: (cheapest unit price) $\times$ (number of units needed)

12.6 Concrete Examples

Example 1: Coffee is \$3, tea is \$5, want utility 10

$$\begin{aligned}e(3, 5, 10) &= \min\{3, 5\} \times 10 \\&= 3 \times 10 \\&= \$30\end{aligned}$$

Why? Coffee is cheaper (\$3 < \$5), so buy 10 coffees at \$3 each = \$30 total.

Example 2: Coffee is \$5, tea is \$2, want utility 10

$$\begin{aligned}e(5, 2, 10) &= \min\{5, 2\} \times 10 \\&= 2 \times 10 \\&= \$20\end{aligned}$$

Why? Tea is cheaper (\$2 < \$5), so buy 10 teas at \$2 each = \$20 total.

12.7 The Big Picture

Understanding the Expenditure Function

What it is: A function that tells you the minimum cost to reach a given utility level.

Inputs:

- p_1, p_2 = prices of the two goods
- u^* = target utility level

Output: The dollar amount you need to spend

How to get it:

1. Find compensated demand (part a): What quantities should you buy?
2. Calculate the cost: price $\times$ quantity for each good, then add them up

For perfect substitutes: $e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^*$

13 Part (c): Understanding Substitution Effects

13.1 What Is the Substitution Effect?

When a price changes, your consumption changes. But WHY does it change? Economists split this into TWO separate reasons:

1. **Substitution Effect:** You buy less of the good because it got relatively more expensive (holding utility constant)
2. **Income Effect:** You buy less because the price increase makes you effectively poorer (can't afford as much)

Part (c) asks ONLY for the substitution effect. This means we ignore the income effect and ask:

"If I wanted to stay at my original happiness level, how would my consumption change?"

Key Formula

Substitution Effect = (New compensated demand) - (Original demand)

Where "new compensated demand" means: the cheapest way to reach the **original utility** with **new prices**.

This is exactly what you calculated in part (a)! That's why the problem says "using your answer to (a)."

13.2 Setup from the Problem

The problem tells us:

- Initially: $p_1 = 1, p_2 = 2, m = 10$
- Consumer buys bundle $(10, 0)$ and earns utility 10
- Utility function: $u(x_1, x_2) = x_1 + x_2$

Why $(10, 0)$? Because good 1 is cheaper (\$1 < \$2), so with perfect substitutes, buy only good 1. With \$10, you can buy 10 units of good 1.

Original utility: $u^* = 10 + 0 = 10$

13.3 Part (c)(i): Price Increases to 1.5

Question: Calculate the substitution effect if p_1 increases to 1.5.

13.3.1 Step 1: Set Up

- **Original situation:** $p_1 = 1, p_2 = 2 \rightarrow$ bought $(10, 0) \rightarrow$ utility = 10
- **New prices:** $p_1 = 1.5, p_2 = 2$
- **Target utility:** Keep $u^* = 10$ (the original utility)

13.3.2 Step 2: Find New Compensated Demand

Use your answer from part (a): what bundle reaches $u^* = 10$ cheaply with new prices?

Check which good is cheaper:

- Good 1 now costs \$1.5
- Good 2 still costs \$2
- Since $1.5 < 2$, good 1 is still cheaper!

From part (a), when $p_1 < p_2$:

$$x_1^c(1.5, 2, 10) = (u^*, 0) = (10, 0)$$

So the new compensated demand is $(10, 0)$ - still buy 10 units of good 1, 0 of good 2.

13.3.3 Step 3: Calculate Substitution Effect

$$\begin{aligned}\text{Substitution Effect} &= (\text{New compensated demand}) - (\text{Original demand}) \\ &= (10, 0) - (10, 0) \\ &= (0, 0)\end{aligned}$$

Answer to Part (c)(i)

Substitution effect:

- Change in good 1: $\Delta x_1 = 0$
- Change in good 2: $\Delta x_2 = 0$

The substitution effect is **zero**!

13.3.4 Why Is It Zero?

Even though good 1 got more expensive (from \$1 to \$1.50), it's **still the cheaper good**. Since goods are perfect substitutes, you still buy only good 1 to minimize cost. There's no substitution to good 2 because good 2 is still more expensive.

Key insight: With perfect substitutes, substitution happens only when the price change makes the other good become cheaper!

13.4 Part (c)(ii): Price Increases to 2.5

Question: Calculate the substitution effect if p_1 increases to 2.5.

13.4.1 Step 1: Set Up

- **Original situation:** $p_1 = 1, p_2 = 2 \rightarrow$ bought $(10, 0) \rightarrow$ utility $= 10$
- **New prices:** $p_1 = 2.5, p_2 = 2$
- **Target utility:** Keep $u^* = 10$

13.4.2 Step 2: Find New Compensated Demand

Check which good is cheaper:

- Good 1 now costs \$2.5
- Good 2 still costs \$2
- Since $2.5 > 2$, good 2 is now cheaper!

From part (a), when $p_1 > p_2$:

$$x_1^c(2.5, 2, 10) = (0, u^*) = (0, 10)$$

So the new compensated demand is $(0, 10)$ - buy 0 of good 1, 10 of good 2.

13.4.3 Step 3: Calculate Substitution Effect

$$\begin{aligned}\text{Substitution Effect} &= (\text{New compensated demand}) - (\text{Original demand}) \\ &= (0, 10) - (10, 0) \\ &= (-10, +10)\end{aligned}$$

Answer to Part (c)(ii)

Substitution effect:

- Change in good 1: $\Delta x_1 = -10$ (decrease by 10 units)
- Change in good 2: $\Delta x_2 = +10$ (increase by 10 units)

You completely substitute away from good 1 toward good 2!

13.4.4 Why Such a Large Effect?

Good 1 was \$1, increased to \$2.5 - that's a huge jump! Now good 1 is MORE expensive than good 2. Since they're perfect substitutes (you like them equally), you switch completely to good 2.

Key insight: With perfect substitutes, the substitution effect is all-or-nothing. You either stay with the same good (if it's still cheaper) or switch completely to the other good (if it becomes cheaper).

13.5 Summary: Comparing Both Cases

Scenario	New p_1	Which is cheaper?	Substitution Effect
Original	1	Good 1 (\$1 ; \$2)	-
Part (c)(i)	1.5	Good 1 (\$1.5 ; \$2)	(0, 0) - no change
Part (c)(ii)	2.5	Good 2 (\$2 ; \$2.5)	(-10, +10) - complete switch

Big Picture

The substitution effect measures how you respond to relative price changes when trying to maintain your original happiness level.

For perfect substitutes:

- If the good that got more expensive is **still cheaper**: No substitution ($\Delta = 0$)
- If the price increase makes it **more expensive than the alternative**: Complete substitution (switch entirely to the other good)

This is why perfect substitutes have "extreme" substitution effects - it's all or nothing!